

Stability vs flexibility: reshaping archaeal membranes in silico

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eLife Assessment

This **important** study characterizes the mechanics and stability of bolalipids from archaeal membranes using molecular dynamics simulations. A mesoscale model of bolalipids is presented and evaluated across a series of membrane configurations. The evidence supporting the conclusions is **convincing**, demonstrating that mixtures of bolalipids and regular bilayer lipids in archaeal membranes enhance fluidity and stability.

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Abstract

Across the tree of life, distinct designs of cellular membranes have evolved. In bacteria and eukaryotes single-headed lipids self-assemble into flexible bilayer membranes. By contrast, archaea often possess double-headed, monolayer spanning bolalipids, mixed with bilayer lipids, enabling them to survive in harsh environments. Here, using a minimal computational model for bolalipid membranes, we discover trade-offs when forming membranes. We find that membranes made out of flexible bolalipids resemble bilayer membranes as bolalipids exhibit conformational switch into U-shaped conformations to enable higher curvatures. Conversely, stiffer bolalipids, resembling those in extremophile archaea, take on straight conformations and form liquid membranes that are stiff, and prone to pore formation during membrane reshaping. Strikingly, we show how to achieve fluid bolalipid membranes that are both stable and flexible – by including small amounts of bilayer lipids, as archaea do. Our study explains how different organisms resolve trade-offs when generating membranes of desired material properties.

Introduction

A bounding membrane separates the interior of every biological cell from its extracellular environment and gives the cell its shape. To perform its function, this bounding membrane must be sturdy and impermeable to the leakage of small charged molecules like ions to allow the generation of an electrochemical gradient [1], but also flexible enough to be remodelled during

essential cellular processes like cell division and vesicle formation [2]. These requirements are partially conflicting and difficult to combine. Interestingly, two different generic membrane designs have evolved across the tree of life to solve this problem [3]. Bacterial and eukaryotic membranes possess fatty acid lipids that have a hydrophilic head group linked to hydrophobic tails via ester-linkages, which self-assemble into bilayer membranes [4]. By contrast, archaeal membranes lipids are constructed from branched isoprenoid lipids. These can include cyclopentane rings and which are attached via an ether linkage to one (e.g., archaeol, **Fig. 1A** left) or two hydrophilic heads, leading them to be called bipolar lipids or simply bolalipids (e.g., caldarchaeol, **Fig. 1A** right) [4]. As an ensemble, bilayer and bolalipids can self-assemble into fluid lipid membranes of different architectures - as bilayer lipid membranes, bolalipid membranes or mixture membranes (**Fig. 1B** right).

As a result of this distinct lipid composition and geometric organisations, archaeal membranes tend to have markedly different properties from eukaryotic and bacterial membranes. This feature of their membrane biochemistry is thought to be responsible, in part, for the ability of some archaea to survive under conditions like those found in volcanic springs at temperatures > 75°C, despite them lacking a cell wall. This hypothesis is supported by the observation that archaea alter their membrane composition depending on the environmental conditions to maintain a fluid but stable membrane [6]. Particularly, it has been found that at higher temperatures, the fraction of bolalipids in archaeal membranes increases [7], and bolalipids contain an increasing number of cyclopentane rings within their lipid tails, which can make the molecules more rigid and prone to aggregation [8]. For instance, the cell membrane of one of the standard archaeal model systems, *Sulfolobus acidocaldarius*, which lives at 80°C and pH 2~, can contain over 90 % of bolalipids [8]. Besides temperature adaption, archaeal lipids also exhibit dense packing, high viscosity, and low porosity to small molecules like protons [9], enabling the *Sulfolobus* membrane to sustain pH gradients of 4 [10]. Intriguingly, some of the membrane-reshaping proteins which originated in archaea, such as ESCRTs [11] can also function with the membranes of eukaryotes, which are believed to have evolved as a merger between bacteria and archaea [12] and which have membranes made of bacterial lipids.

All this makes the study of reshaping archaeal membranes extremely interesting. Unfortunately, it is fairly challenging to investigate archaeal cells experimentally due to their unusual chemistry and extreme living conditions [13]. Therefore, most of what we know about archaeal tetraether lipid membranes thus far has been collected from studying *in vitro* reconstituted membranes. For instance, the conformation of individual lipids in bolalipid membranes was studied at the water-air interface [14, 15] or using NMR experiments on lipid vesicles [16]. This suggested the existence of U-shaped lipid conformations in archaeal type membranes (**Fig. 1B**). Moreover, *in vitro* reconstituted vesicles primarily composed of bolalipids can fuse with influenza virus particles at similar kinetic rates compared to bilayer vesicles, further suggesting that bolalipids exist in U-shape allowing for membrane remodeling and fusion [17]. Membrane properties like bending stiffness [18] or lipid phase [19] have also been measured in vesicles prepared from archaeal tetraether lipids to demonstrate that archaeal lipid derived membranes are unique in being stable up to temperatures of 80°C [19]. Moreover, experiments with lipid vesicles made from synthetic bilayer lipids that include cyclopentane rings, which naturally appear in lipids of extremophilic archaea, showed that increasing the number of these rings increases membrane rigidity [20].

From the view point of membrane physics, the remodeling of bilayer membranes has been studied for decades using continuum models and computer simulations [21–23]. By contrast, apart from the aforementioned experimental studies and a few studies either using finegrained MD simulations or theory [24–26], little research has been devoted to investigating the properties of archaeal bolalipid membranes despite the obvious importance of this question from

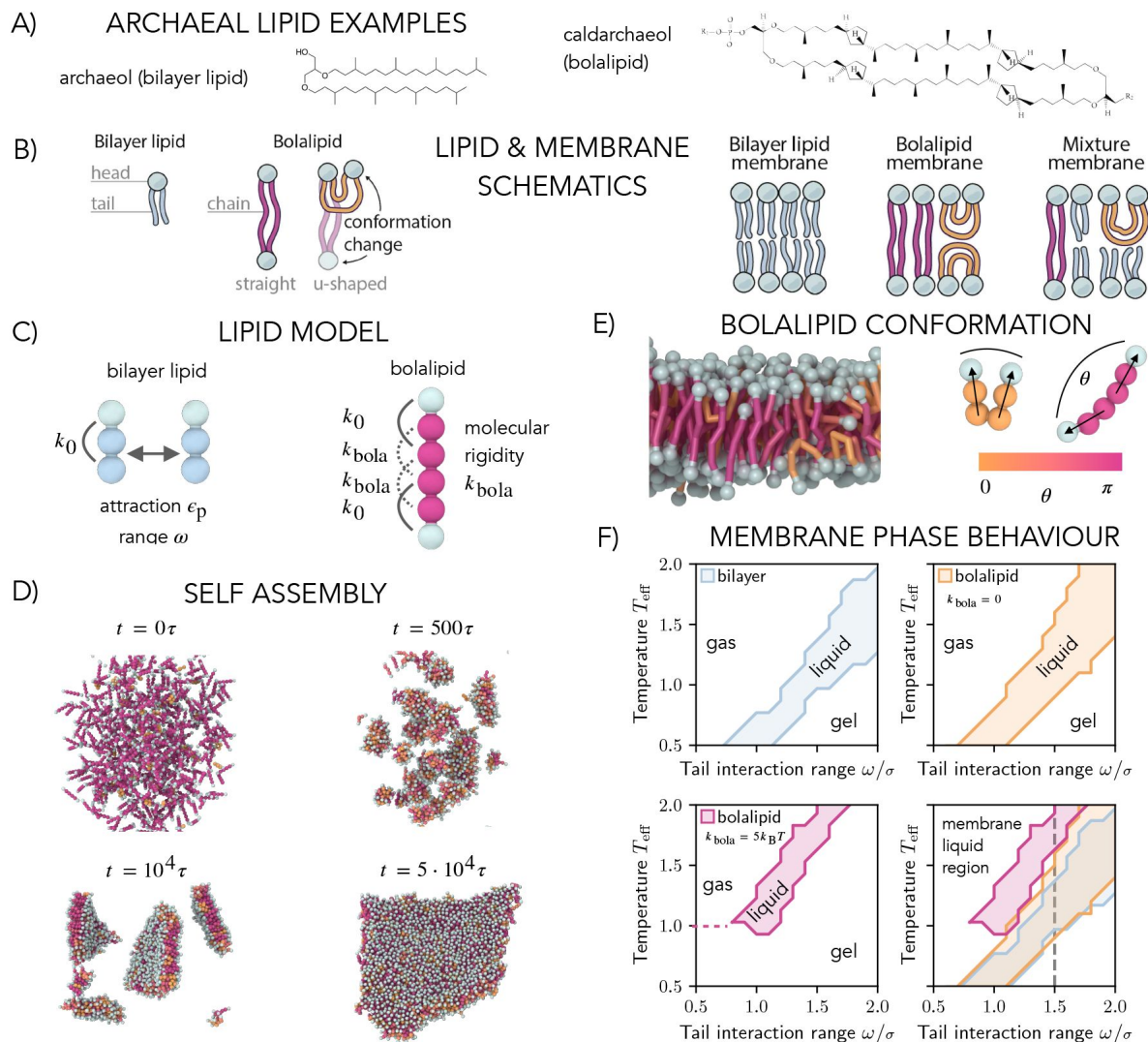


Figure 1.

Computational model and phase space of bilayer and bolalipid membranes.

(A) Structure of the diether bilayer lipid archaeol (left) and the tetraether bolalipid caldarchaeol including four cyclopentane rings (right), both present in the membrane of *Sulfolobus acidocaldarius*, a common archaeal model system that lives at high temperatures and low pH [5]. (B) Schematics for a bilayer lipid, a bolalipid and its two in-membrane conformations, membranes made of bilayer molecules only, bolalipid molecules only, and a mixture of the two (left to right). (C) Bilayer lipid (left) is described with one head bead and two tail beads straightened by an angular potential of strength k_0 . Tail beads of different lipids attract with the strength ϵ_p and the range ω . Bolalipids (right) consist of two bilayer lipids connected by a bond and straightened by an angular potential of strength k_{bola} . (D) Snapshots of bolalipids self-assembling into a flat membrane ($k_{bola} = 0.3 k_B T$). (E) Cross-section of self-assembled membrane (right), with bolalipids colored according to their conformation: straight lipids in crimson and U-shaped lipids in orange. (F) Membrane phase behavior: liquid, gel and gas regions as a function of the effective temperature T_{eff} and tail interaction range ω for bilayer (top left) and membranes made of flexible $k_{bola} = 0$ (top right) and stiff $k_{bola} = 5 k_B T$ (bottom left) bolalipid molecules. Overlays of all liquid regions (bottom right) show that stiffer lipids exhibit fluid membrane region at higher temperatures. The dashed line marks $\omega = 1.5\sigma$, the value we used in the rest of the work.

evolutionary, biophysics, and biotechnological perspectives. Particularly, membrane reshaping at the mesoscale has been largely neglected although different membrane physics are expected to manifest.

To address this question, here we have designed a minimal model of archaeal membranes that can be used to explore the impact of both bolalipids and bilayer lipids on membrane biophysics and membrane reshaping. Our coarse-grained molecular dynamics simulations show that the geometry of bolalipids is, through the effects of entropy alone, sufficient to shift the fluid phase of archaeal-type membranes so that they are stable at high temperatures. In addition, we show that membranes assembled from bolalipids can have a much higher bending rigidity than bilayer-derived membranes, and are more likely to resist membrane shape changes as in fission. During membrane deformation, stress in these bolalipid membranes is relieved via a small fraction of bolalipids taking up a U-shaped conformation, which renders them a mechanically switchable material, or by the formation of large pores. Remarkably, however, when combined, a mixture of bilayer and bolalipids generates membranes that are stable at high temperature, which becomes softer as the fraction of bilayer lipids increases. Taken together, our results demonstrate how doping a bolalipid membrane with a small fraction of bilayer lipids can relieve the trade-off, enabling membranes to be both stable under extreme conditions and to be remodelled without leaking.

Results

Computational Model

To study bolalipid membranes and compare them to bilayers, we extended the Cooke and Deserno model for bilayer membranes [27]. In the original model for the bilayer a single bilayer lipid is represented by a chain of three nearly equally sized beads of diameter of $\sim 1\sigma$, where σ is our distance unit and roughly maps to 1 nm (**Fig. 1C** left); one bead stands for the head group (cyan) while the others represent the hydrophobic tail (blue). Each adjacent pair of beads in a lipid is linked by a finite extensible nonlinear elastic (FENE) bond. The angle formed by the chain of three beads is kept near 180° via an angular potential with strength k_0 . While lipid heads interact exclusively through volume exclusion, the beads of lipid tails interact via a soft attractive potential of the strength ϵ_p and range ω (**Fig. 1C** left), effectively modelling hydrophobic interaction in an implicit solvent. This interaction strength governs the membrane phase behavior and can be interpreted as the effective temperature $T_{\text{eff}} = k_B T / \epsilon_p$.

To model a bolalipid molecule, we joined two bilayer lipids so that a lipid molecule is formed with a head bead (cyan) that is linked to four tail beads (crimson) which are again linked to another head bead (**Fig. 1C** right). In this way, both bilayer lipids and bolalipids share the same molecular structure and the same interactions between lipid beads. Bolalipids in archaeal membranes can differ in the number of cyclopentane rings or the branching of the tail and thus in the molecular stiffness [6, 28]. To represent this effect, we added two angular potentials between the second and the fourth and the third and the fifth tail bead with variable strength k_{bola} . By varying k_{bola} , we can control the molecular stiffness of the bolalipid molecules and thus model different types of bolalipids. The model is simulated within molecular dynamics implemented in the LAMMPS open source package [29], and simulations are visualized with OVITO [30]. To include the implicit effect of the surrounding water and to simulate membranes at vanishing tension, we used a Langevin thermostat combined with a barostat that kept the membrane at zero pressure in the x - y plane. Details on the computational model are given in the Supplemental Information (SI) section 1.

Self-assembly and phase behavior

We first tested the ability of both the bilayer and bolalipid molecules to self-assemble into membranes. To do so, we placed dispersed lipids in a periodic 3D box. For both bilayer and bolalipids we found that membranes self-assembled over a wide range of lipid interaction parameters (Fig. 1D, SI section 2, and Movie S1). We then explored the influence of flexibility in the bolalipid molecule. At small values of the molecular rigidity k_{bola} , single lipids are flexible and can thus adopt a range of possible conformations. The different conformations can be classified by the angle θ between the two lipid heads (Fig. 1E). Two lipid conformations dominated the conformation distribution in the context of a membrane: the U-shaped conformation with both head beads on the same membrane leaflet ($\theta \approx 0$) and the straight conformation with one head bead in each opposing membrane leaflet ($\theta \approx \pi$) (Movie S2.b, Fig. S1, and SI section 3; for comparison, bilayer membranes in Movie S2.a and rigid bolalipid membranes in Movie S2.c). In the self-assembled bolalipid membrane, we marked bolalipids as being in the U-shape conformation if $\theta < \pi/2$ and in the straight conformation otherwise.

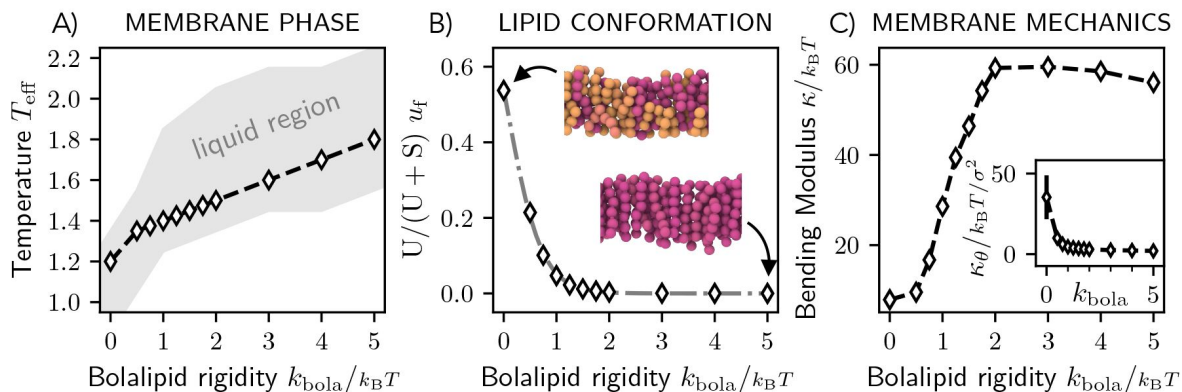
To study the phase behavior of bilayer and bolalipid membranes, we analyzed the diffusion of single lipids in self-assembled membranes as a function of the lipid interaction parameters ϵ_p and ω (SI sections 4 and 5). The diffusion constant D exhibited a discontinuity as a function of the temperature, which marks the transition as the membrane moves from the gel phase to the liquid phase. The discontinuity occurred at different values of interaction strength ϵ_p and interaction range ω for bilayer and bolalipid lipids, and it also depended on the values of the molecular stiffness k_{bola} (Fig. S4). In these simulations, the disintegration of the membrane defined an upper limit to the liquid phase and the transition to the gas phase. Based on this classification, we plotted the phase diagram for bilayer membranes (Fig. 1F top left), fully flexible bolalipid membranes ($k_{\text{bola}} = 0$, Fig. 1F top right), and rigid bolalipid membranes ($k_{\text{bola}} = 5 k_B T$, Fig. 1F bottom left) as a function of the range of the hydrophobic interaction ω and the temperature T_{eff} . Membranes made of bilayers (blue) and flexible bolalipid molecules ($k_{\text{bola}} = 0$, orange) behaved similarly under these conditions (Fig. 1F bottom right). Strikingly, just as observed in extremophile archaea, as the molecular stiffness of bolalipids increased ($k_{\text{bola}} = 5 k_B T$, magenta), the liquid region was shifted toward higher temperatures and larger values of the interaction range. This is due to the fact that bolalipid molecules are able to engage in more extensive interactions with partners when in the extended conformation, which helps to stabilize the membrane at higher temperatures.

Bolalipid conformations and mechanical properties

To explore mechanical properties of bolalipid membranes, we chose $\omega = 1.5 \sigma$ for the remainder of the work (dashed line in Fig. 1F bottom right). Fig. 2A shows the phase diagram for bolalipid membranes replotted as a function of temperature and bolalipid rigidity for the chosen interaction range. To be able to compare membranes made of bolalipids of different molecular rigidities, we needed to adjust temperature for each to reach similar fluidities, shown by dashed lines in Fig. 2A. We then characterised the conformations of individual bolalipids in flat membranes, measured by the fraction of bolalipids in U-shape conformation u_f . We find that for flexible bolalipids ($k_{\text{bola}} = 0$), more than 50% of all lipids are $\geq$ in the U-shape conformation (Fig. 2B). This fraction decreases with increasing rigidity of bolalipid molecules and vanishes around $k_{\text{bola}} \geq 2 k_B T$, for which almost all bolalipids take up linear conformations.

This behavior can be easily captured by considering bolalipids as a two state system, with straight and U-shape conformation, as argued before (Fig. S1). We assumed that the energy of the straight conformation vanishes $E_s = 0$ and the energy of the U-shape conformation reads $E_u = c_0 + c_1 k_{\text{bola}}$, where, for simplicity, we assumed that the energy linearly depends on k_{bola} which is the only relevant energy scale in the system and the two constants c_0 and c_1 , which determine the conformation energy of U-shaped bolalipids. The fraction of bolalipids in U-shape conformation

LOW CURVATURE : FLAT MEMBRANES



HIGH CURVATURE : MEMBRANE TETHERS

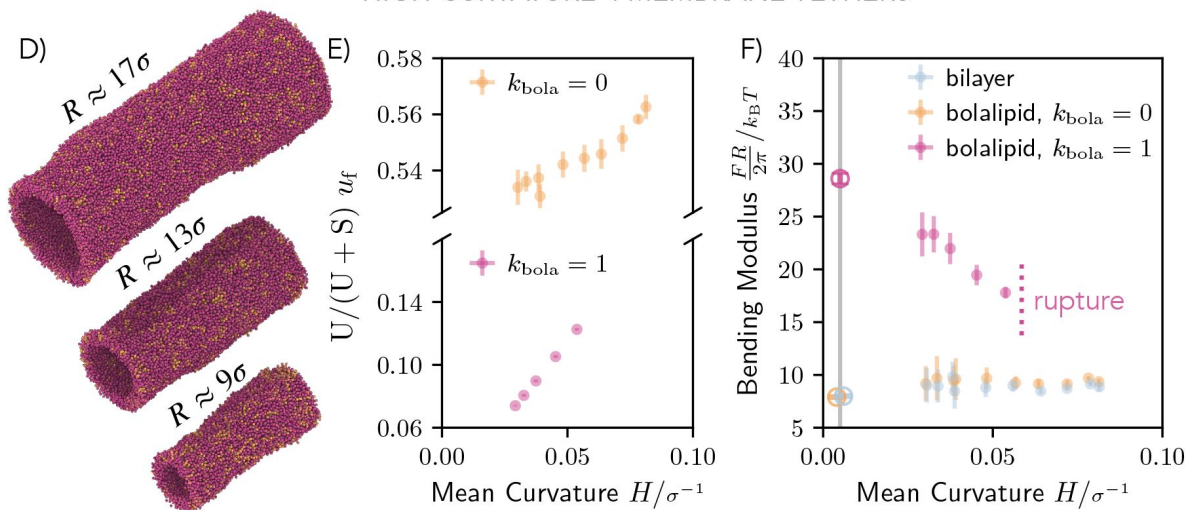


Figure 2.

Mechanics of pure bolalipid membranes.

(A) Liquid region as a function of temperature and bolalipid rigidity for pure bolalipid membranes (gray). The dashed line shows the bolalipid membranes of approximately same fluidities. (B) Fraction of bolalipids in the U-shape conformation ($\theta = 0$), fitted to $u_f(k_{\text{bola}}) = 1/(1 + \exp(\beta(-0.16 + 3k_{\text{bola}})))$ (gray dashed line) according to a two-state model. Insets: simulation snapshots with bolalipids colored according to their conformations. (C) Bending modulus as a function of bolalipid molecule rigidity k_{bola} . Inset: Tilt modulus as a function of bolalipid rigidity k_{bola} . (D) Snapshots of bolalipid membranes at the range of explored curvatures for $k_{\text{bola}} = 1k_{\text{B}}T$. (E) Fraction of bolalipid molecules in the U-shaped conformation as a function of the mean membrane curvature $H = 1/(2R)$ for membranes made of flexible ($k_{\text{bola}} = 0$) and semi-flexible ($k_{\text{bola}} = 1k_{\text{B}}T$) bolalipid molecules. (F) Bending modulus as a function of curvature. For the flat membrane ($H \sim 0$), the corresponding bending rigidity from (C) is marked by the vertical line and empty circles.

then follows $u_f(k_{\text{bola}}) = 1/(1 + \exp(\beta(c_0 + c_1 k_{\text{bola}})))$, with $\beta = 1/(k_B T)$, as shown by the fit in **Fig. 2B** (dashed gray line, $R^2 = 0.99$, see SI section 6). For the fit it appears that $c_0 < 0$, which implies that bolalipids in U-shape conformation are slightly favored over straight bolalipids at $k_{\text{bola}} = 0$ likely because of entropic reasons.

Bolalipid conformations and membrane rigidity

It was previously hypothesized that an increasing fraction of bolalipids in straight configuration would increase the membrane rigidity [18]. To determine the membrane rigidity using our model, we assessed the height fluctuation spectrum (h^2) of flat membranes in a periodic box. [27, 31]. Interestingly, we found that the original theory of Helfrich [32] failed to describe the resulting height fluctuation spectrum. However, the extended theory by Hamm and Kozlov [33], which also includes the energetic cost of lipid tilt, successfully captured bolalipid fluctuations. In this case, the resulting height spectrum of the membrane at vanishing membrane tension is given by [31]

$$\begin{aligned} \langle |h_n|^2 \rangle &= \frac{k_B T}{L^2} \left(\frac{1}{\kappa q^4} + \frac{1}{\kappa_\theta q^2} \right) \\ &= \frac{k_B T}{L^2} \frac{1}{\kappa} \left(\frac{1}{q^4} + \frac{l_\theta^2}{q^2} \right), \end{aligned} \quad (1)$$

where $q = 2\pi n/L$ is the wave number, L is the box size, κ is the bending rigidity of the membrane, κ_θ is the tilt modulus and $l_\theta = \sqrt{\kappa/\kappa_\theta}$ is a characteristic length scale related to tilt. Considering Eq. (1), the tilt term is expected to matter if the analyzed inverse wave numbers become similar to l_θ . The wave numbers that we analyzed correspond to wavelengths that are at least twice the thickness of the membrane ($q < 2\pi/(12\sigma) \approx 0.5 \sigma^{-1}$). Thus, the tilt term is expected to contribute if $l_\theta > 2\sigma$. For typical bilayer membranes, one finds $\kappa_\theta = 12 k_B T \text{ nm}^{-2}$ [31] and $\kappa = 20 k_B T$ [34], so $l_\theta \sim 1 \text{ nm} \approx 1 \sigma$ and therefore the tilt term can be neglected as practiced before [27]. However, when the membrane rigidity increases as we expect for bolalipid membranes, l_θ increases and the tilt term in Eq. (1) becomes relevant.

By fitting the height spectrum for bolalipid membranes (and bilayer membranes for comparison) (Fig. S6), we measured the bending rigidity (**Fig. 2C**) and tilt modulus (**Fig. 2C** inset) as a function of k_{bola} (see SI section 7 and Movie S3). With increasing bolalipid molecular rigidity k_{bola} , the bending rigidity κ rose from $8 k_B T$ and plateaued at $60 k_B T$, showing bolalipid membranes can be very rigid while liquid. Strikingly, the increase in membrane rigidity coincided with U-shaped bolalipids vanishing from the membrane (**Fig. 2B**), which confirmed the hypothesis that straight bolalipids render lipid membranes rigid. At the same time, the tilt modulus κ_θ decreased with bolalipid rigidity, from $30 \pm 10 k_B T / \sigma^2$ to $2 k_B T / \sigma^2$, lowering less than $1 k_B T$ for $k_{\text{bola}} \geq 2 k_B T$. Since membrane bending and lipid tilting are two modes of membrane deformations that compete, we conclude that bilayer and flexible bolalipids molecules form flexible membranes that prefer to bend rather than to tilt, while bolalipids in straight configuration form rigid membranes that prefer to tilt rather than to bend. Taken together, bolalipid membranes made of flexible lipid molecules are as flexible as lipid bilayers, adopting U-shaped conformations, where those made of bolalipids in straight configurations are rigid.

Interplay between membrane curvature and rigidity

Following geometric intuition, we expect the fraction of molecules in the U-shape to change as the membrane curvature changes, to enable area difference between the two membrane leaflets needed to adapt to the curvature. This would imply that the bending rigidity, importantly, through the fraction of U-shaped bolalipids, is curvature dependent. To investigate the impact, we measured the bending rigidity of bolalipid membranes in cylindrical shapes as a function of their radii [35] (see **Fig. 2D**, Movie S4, and SI section 8). We first noticed that while membrane tubes made of bilayer and flexible bolalipids were stable up to small cylinder radii R , almost as small as the membrane thickness itself, we found that membrane made from stiffer bolalipids

($k_{\text{bola}} = 1k_{\text{B}}T$) ruptured well before. Strikingly, while for stiffer bolalipids ($k_{\text{bola}} = 1k_{\text{B}}T$) the U-shaped bolalipid fraction increased strongly over a short range of the mean membrane curvature ($H = 1/(2R)$), we only found a small change in the U-shaped bolalipid fraction of flexible bolalipid membranes ($k_{\text{bola}} = 0$) (Fig. 2E). Consequently, since the fraction of U-shaped bolalipid molecules controls the bending rigidity of bolalipid membranes (Figs. 2B and 2C), we found that there is a strong dependency of the bending rigidity κ on the membrane mean curvature of stiffer bolalipids ($k_{\text{bola}} = 1k_{\text{B}}T$) (Fig. 2F). In contrast, we did not find that κ was curvature-dependent for bilayer or flexible bolalipid membranes ($k_{\text{bola}} = 0$). Taken together, the rigidity of bolalipid membranes is not only controlled by the molecular stiffness of their lipid constituents but also by the emerging geometry of the ensemble of lipids. Since membrane geometry and thus membrane rigidity will change upon membrane deformations this gives rise to plastic material properties.

Gaussian rigidity of bolalipid membranes

Another important material parameter is the Gaussian bending modulus $\bar{\kappa}$, which characterizes the reshaping behavior of fluid lipid membranes under topological changes [34]. $\bar{\kappa}$ is notoriously difficult to measure since it only becomes detectable when the membrane changes its topological state. Continuum membrane theory, combining stability arguments and elasticity, predicts $-\bar{\kappa}/\kappa \in [-0.5, -1]$ [34], where the former value is expected for incompressible membranes. Indeed, most of the numbers we know for the ratio of the two bending rigidities, many of which were deduced from simulations, lie within this range [36]. Using the same method as developed by Hu et al. [36], we determined $\bar{\kappa}$ by measuring the closing efficiency of membrane patches into a sphere (see SI section 9). We obtained $\bar{\kappa} = -4.61 \pm 1.91 k_{\text{B}}T$ and thus a ratio of $-\bar{\kappa}/\kappa = 1.07 \pm 0.50$ for bilayer membranes.

In contrast, we got significantly larger values (less negative) with $\bar{\kappa} = -2.81 \pm 2.41 k_{\text{B}}T$ and $-\bar{\kappa}/\kappa = 0.55 \pm 0.49$ for flexible bolalipid membranes ($k_{\text{bola}} = 0$). The result shows that in addition to the differences in bending rigidities, the ratio of the two bending moduli differs strongly in bilayer and bolalipid membranes.

Archaeal membranes made of mixtures of bolalipids and bilayer-forming lipids

Archaeal membranes contain varying amounts of bilayer lipids [8, 37]. The exact bolalipid/bilayer fraction depends on the growth temperature, with higher levels of bolalipids with increasing temperature [7], and higher fraction of cyclopentane rings in the tails [28]. In order to investigate the effect of different lipid contents on membrane mechanical properties, we wanted to model the archaeal membrane by mixing bilayer lipids into bolalipid membranes. Since in our model, the liquid regions of rigid bolalipid membranes and bilayer membranes do not overlap (Fig. 1F bottom right), we picked the temperature $T_{\text{eff}} = 1.3$ to minimize fluidity mismatch and we set the molecular rigidity $k_{\text{bola}} = 2 k_{\text{B}}T$ to limit U-shaped bolalipids (Fig. 2B). We then measured the diffusion constant D as a function of the fraction of bilayer lipids f^{bi} (see SI section 5). Interestingly, we found that mixing only 10% bilayer lipids into the bolalipid membrane in gel state is enough to fluidize the membrane (Fig. 3A). We then measured the bending rigidity and the tilt modulus of flat mixture membranes by analyzing the fluctuation spectrum. While the bending rigidity κ decreased (Fig. 3B), the tilt modulus increased non-linearly with the bilayer lipid fraction f^{bi} (Fig. 3B inset). Taken together, the bolalipid membrane can be substantially softened either through bolalipids acquiring U-shaped conformation or through addition of bilayer-forming lipids.

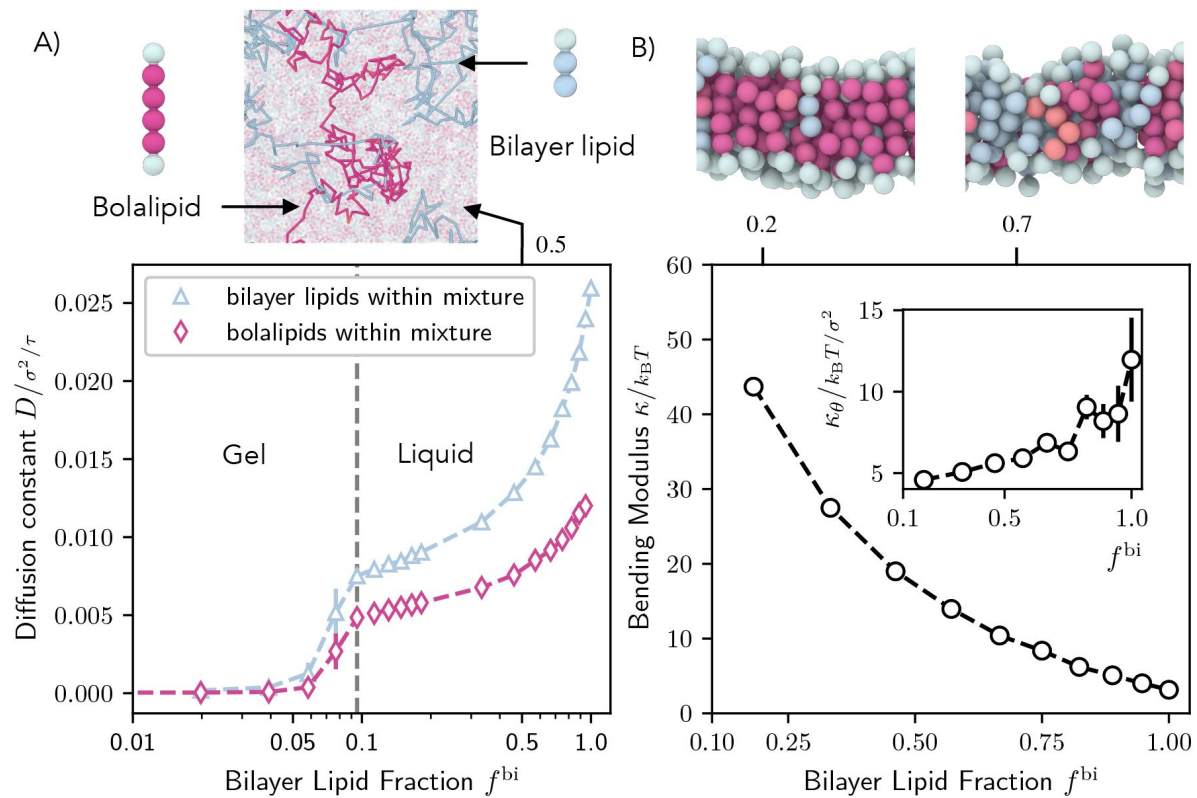


Figure 3.

Fluidity and rigidity of mixed bilayer/bolalipid membranes.

(A) Single lipid diffusion constant for each species as a function of bilayer fraction f^{bi} ($k_{bola} = 2k_B T$, $T_{eff} = 1.3$). For $f^{bi} \geq 0.1$, the resulting mixture becomes liquid. Top: Diffusion trajectories of a bolalipid (blue) and a bilayer lipid (red) in a mixture membrane at $f^{bi} = 0.5$. (B) Bending rigidity κ and (Inset) tilt modulus κ_θ as a function of the fraction of bilayer molecules f^{bi} . Top: Snapshots show bilayer lipids (blue) in mixed membranes at two different values of f^{bi} .

Curving bolalipid membranes

To investigate the response of bolalipid membranes to large membrane curvature and topology changes like those induced upon vesicle budding, which regularly occurs in archaea, we simulated membrane wrapping of an adhesive cargo bead (**Fig. 4A** [↗](#), Movies S5.a to S5.c, and SI section 10). Importantly, this provided us with a method to study how lipid organization is affected by externally imposed membrane curvature and mechanics. We first simulated membrane wrapping at different adsorption energies ϵ_{mc} between lipid head beads and the cargo until we observed that the membrane wrapped the cargo completely (including membrane fission). Then the minimum adsorption energy, for which a membrane bud completely enveloped the cargo bead, is the onset adsorption energy ϵ_{mc}^* (**Fig. 4A** [↗](#)), which we measure as a function of the bolalipid stiffness k_{bola} (**Fig. 4B** [↗](#)). For small molecular stiffness k_{bola} , ϵ_{mc}^* first increases linearly with k_{bola} before it saturates around $k_{bola} = 3 k_B T$. We expect that the onset energy is proportional to the membrane bending rigidity $\epsilon_{mc}^* \propto \kappa$, because the bending energy to wrap a spherical particle is size-invariant [34 [↗](#), 38 [↗](#)]. When we increased the bending rigidity, through increasing stiffness of bolalipid molecule k_{bola} , ϵ_{mc}^* increased by a factor of 3 (**Fig. 4B** [↗](#)), suggesting that also κ increased by a factor of 3. However, from directly measuring the membrane rigidity from the fluctuation spectrum (**Fig. 2C** [↗](#)), we saw that κ increased by a factor of 10. To reconcile these seemingly conflicting observations we reason that the bending rigidity κ , similar to **Fig. 2F** [↗](#), is not constant but softens upon increasing membrane curvature, due to dynamic change in the ratio between bolalipids in straight and U-shaped conformation. Hence, bolalipid membranes show stroking plastic behavior as they soften during reshaping.

Through analysing the bolalipid conformations, we found that the membrane was able to curve by increasing the fraction of U-shaped bolalipids in the outer layer of the deformation (**Fig. 4C** [↗](#) and SI section 12). To a lesser degree, we also observed this effect on the inner membrane neck (**Fig. S12**). Remarkably, though, even when lipids are so stiff that there are no more U-shaped bolalipids in the flat mother membrane ($k_{bola} = 2 k_B T$), the outer layer of the a curved membrane retained a non-negligible fraction of $\approx 10\%$ U-shaped bolalipids, which in turn decreases κ and softens the membrane. However, the softening effect on the membrane, indicated through a constant onset energy for $k_{bola} \geq 3 k_B T$ (**Fig. 4B** [↗](#)), persists even for those very stiff bolalipids. Since for stiff membranes, practically all U-shaped bolalipids are gone (**Fig. 4C** [↗](#)), this suggested that an additional membrane-curving mechanism must be involved.

Looking more closely, at high molecular rigidity ($k_{bola} \geq 2 k_B T$) we observed the formation of multiple pores on the membrane bud, which we quantified by measuring the time-averaged maximum pore diameter (**Fig. 4D** [↗](#), see SI section 12 and Movie S6). While large pores were not observed in the flat membrane, the diameter of membrane pores around the cargo was found to grow with the increase in bolalipid stiffness. We reasoned that pores form when the energetic cost required to change the bolalipid conformation to release bending stress is larger than the energetic cost of opening a lipid edge surrounding the pore. Hence, for relatively flexible bolalipids, U-shaped bolalipids provide the necessary area difference between the outer and inner layer of the membrane bud and thereby soften the membrane. For stiff bolalipid molecules, however, membrane pores start to form to enable membrane curvature as U-shaped bolalipids become prohibited. Both mechanisms help to explain the discrepancy between ϵ_{mc}^* and the bending modulus κ obtained by studying membrane fluctuations (**Fig. 2C** [↗](#)).

Curving archaeal membranes

Having shown that bolalipid membranes can effectively soften also by including some amount of bilayer-forming lipid molecules, we next measured the onset energy ϵ_{mc}^* for cargo budding in the membranes formed by mixtures of bolalipids and bilayer-forming lipids, as a function of bilayer lipid head fraction f_{hi}^{bi} (**Fig. 5A** [↗](#) and Movie S5.d). We found that the onset energy sharply

PURE BOLALIPID MEMBRANES

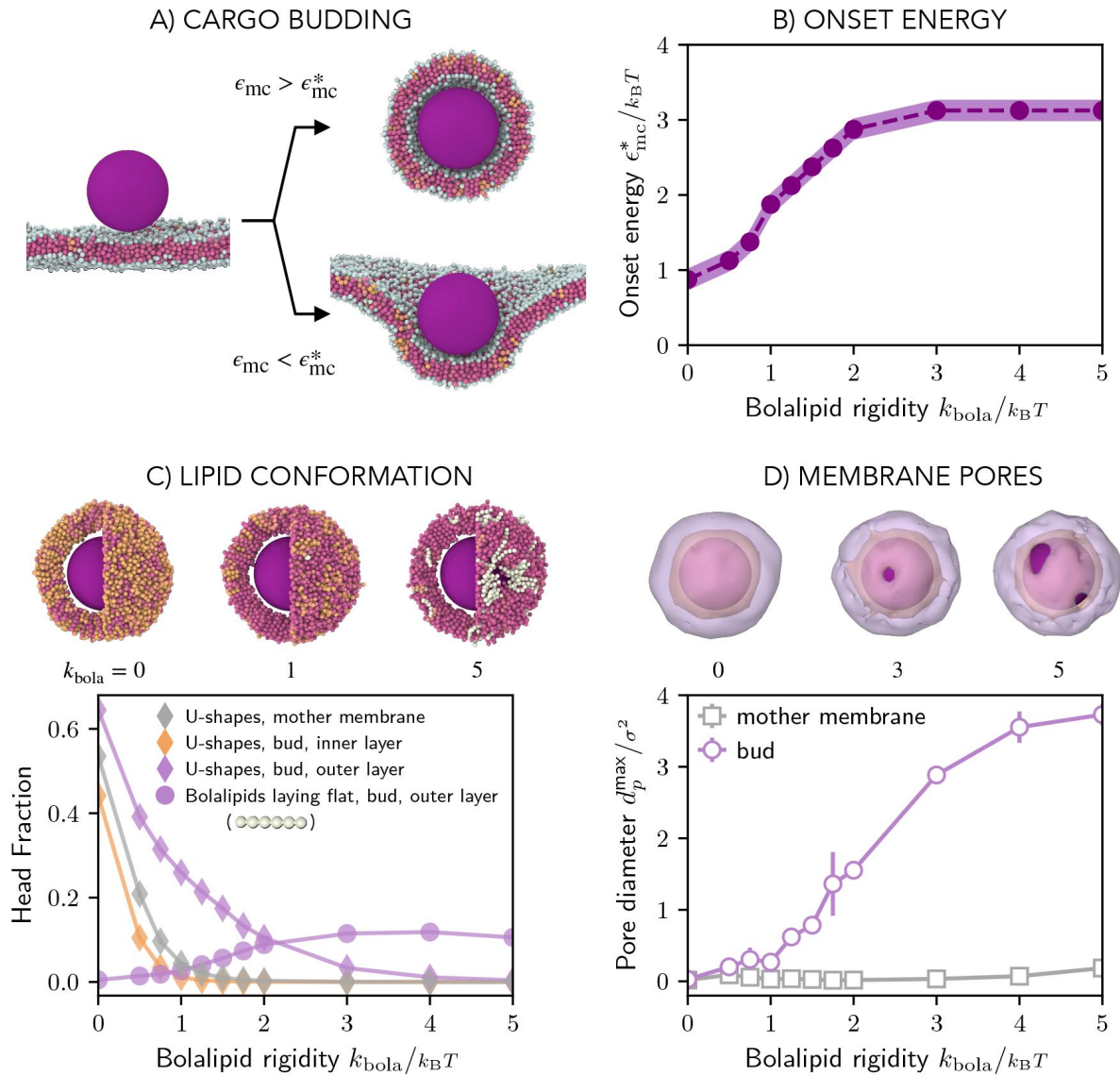


Figure 4.

Reshaping of pure bolalipid membranes.

(A) Simulation snapshots of the membrane wrapping a cargo bead adsorbing onto it. Above the onset adsorption energy ϵ_{mc}^* , the cargo is fully wrapped by the membrane and buds off the mother membrane. (B) Onset energy ϵ_{mc}^* as function of the bolalipid molecule rigidity k_{bola} (for the parameters defined by the line in Fig. 2A). (C) Bottom: Fraction of bolalipids in the U-shape conformation u_f in the outer and inner layers of the membrane bud, and in the flat mother membrane, as function of the bolalipid molecule rigidity k_{bola} . Top: Snapshots and cross-sections of the membrane around the cargo bud. At high bolalipid rigidity the pores form around the cargo, and are lined with bolalipid molecules lying flat around the pore in a straight conformation, with both heads in the outer layer (colored in white). The rest of bolalipids colored according to their head-to-head angle as before. (D) Bottom: Average diameter of transient pores in the membrane bud and the mother membrane as function of the bolalipid molecule rigidity k_{bola} . Pores are defined as membrane openings through which a sphere of diameter 1σ can cross. Top: Snapshots of the membrane surface with outer and inner leaflet surface colored in purple and orange, respectively, intersecting at the rim of the pore (gray).

decreases with increasing amount of bilayer forming lipids, and plateaus for 50% bilayer head fraction, where it acquires similar values as in the case of fully-flexible bolalipids (**Fig. 4B**). For small bilayer fractions, U-shaped bolalipids localize almost exclusively on the outer layer of the bud (**Fig. 5B**). As the bilayer fraction increases, there is a steady reduction in the percentage of U-shaped bolalipids in the outer layer in favor of bilayer lipids that take their role in supporting membrane curvature, with U-shaped bolalipids completely vanishing at high fractions of bilayer-forming lipids. A fraction of bilayer lipid head beads initially shows an asymmetry between the preferred outer layer and the penalized inner layer around the cargo (**Fig. 5C**), but eventually approaches f_h^{bi} in both layers. Taken together, as the bilayer lipid fraction increases, the role of U-shaped bolalipids in making up the asymmetry between the outer and inner layer, is taken over by bilayer lipids. Curiously, the addition of bilayer lipids promotes the formation of U-shaped lipids, both in the flat membrane and even in the inner layer around the bud. This is likely to be explained by the fact that when more bilayer lipids are incorporated, the membrane is less densely packed (**Fig. S10B**) and thus U-shaped bolalipids are promoted.

Importantly, we observed nearly no pores in the membrane bud in mixed membranes, even when we only had very little fraction of bilayer-forming lipids (**Fig. 5D**). Only as the bilayer fraction increased, we observed the formation of very small pores in the bud. For the flat mother membrane, however, membrane pores started to form with increasing values of f^{bi} . They acquired sizes similar to those obtained around the bud in pure bolalipid membranes (**Fig. 4D**). The pore formation in the flat mother membrane is likely promoted because the membrane becomes destabilized by the increasing proportion of bilayer lipids which are close to the gas phase. Taken together, bolalipids can bud porelessly when bilayer-forming lipids, which cause membrane softening, are included.

Discussion

All biological cells are enclosed in fluid lipid membranes. These must be both sturdy and flexible to contain the cellular interior and allow membrane reshaping during division and vesicle formation. Across nature, mono- and bilayer membranes, composed of bilayer and bolalipids have evolved that allow to fulfill these partially conflicting requirements. In this work, we have developed the first minimal model that allows us to transparently compare the distinct behavior of bilayer and bolalipid membranes. In our model, bolalipids are formed by joining together two bilayer lipids, with an adjustable molecular stiffness at the hinge point. Using our model we find striking differences between bilayer and bolalipid membranes in terms of stability, rigidity and plasticity. Our results highlight how bolalipid membrane behave effectively as two-component systems, exchanging bolalipids between different conformations and thereby allowing the material properties to be adapted to support membrane reshaping.

While flexible bolalipid membranes are liquid under the same conditions as bilayer membranes, we found that stiff bolalipids form membranes that operate in the liquid regime at higher temperatures. These results agree well with previous molecular dynamics simulations that suggested that bolalipid membranes are more ordered and have a reduced diffusivity compared to bilayer membranes [24, 25]. In our simulations, this is due to the fact that completely flexible bolalipids molecules adopt both straight (transmembrane) as well as the U-shaped (loop) conformation with approximately the same frequency. In contrast, stiff bolalipids typically only take on the straight conformation when assembled in a membrane. These results agree with the previous coarse-grained molecular dynamics simulations using the MARTINI force field which showed that the fraction of straight to U-shaped bolalipids increased upon stiffening the linker between the lipid tails [24].

MIXTURE OF BOLALIPIDS AND BILAYER-FORMING LIPIDS

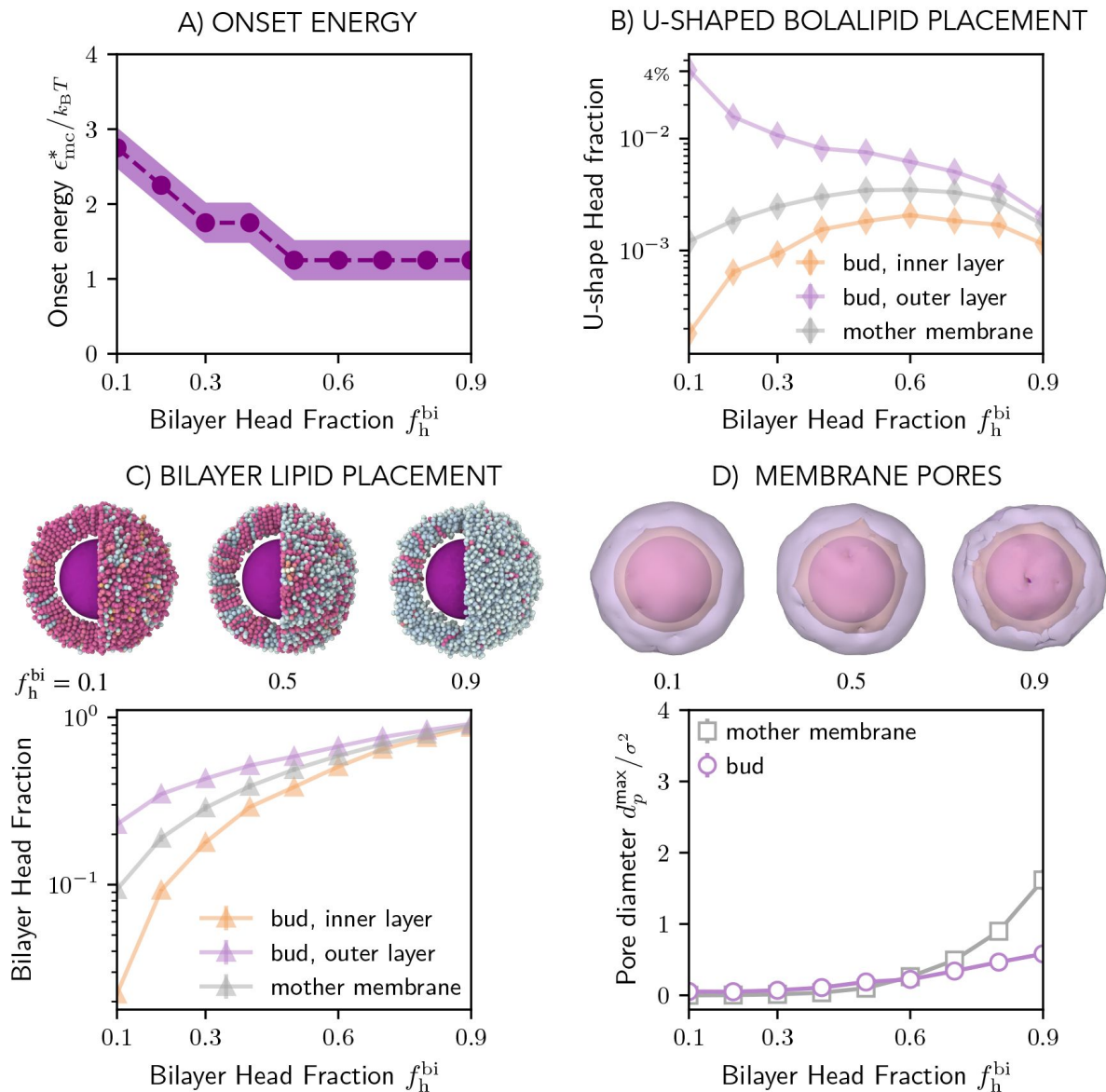


Figure 5.

Curving of the mixed membranes, made of bilayer and bolalipid molecules.

(A) Onset energy required to form the membrane bud, ϵ_{mc}^* , as function of bilayer head fraction f_h^{bi} (for the parameters defined in **Fig. 3**). (B and C) Fraction of U-shaped bolalipid molecules u_f (B) and bilayer molecules (C) in the outer and inner layers of the membrane bud and in the flat mother membrane as a function of the bilayer head fraction f_h^{bi} . Top panels show the respective snapshots of membrane surface around cargo, where bilayer lipids are shown in light blue as in **Fig. 4**. (D) Average diameter of transient pores in the membrane bud and the mother membrane as function of bilayer head fraction f_h^{bi} and respective snapshots of membrane leaflet surfaces surrounding the bud (Top panel).

When we determined the bending rigidity of bolalipid membranes by measuring their response to thermal fluctuations, we found that membranes made from flexible bolalipids are only slightly more rigid than bilayer membranes. This result is consistent with previous atomistic simulations, which showed that the membrane rigidity was similar for membranes composed of bilayer lipids and flexible synthetic bolalipids [39]. Moreover, the result is consistent with a continuum theory which predicted that the rigidity of membranes formed of triblock copolymers is 20% larger than that of diblock copolymers [40]. However, bolalipids in extremophilic archaea are not predicted to be fully flexible as they are expected to pack tighter due to a large number of cyclopentane rings in the lipid tails [6, 28]. Indeed, we found that membranes made of stiff bolalipid molecules can exhibit stiffness that is more than an order of magnitude larger than that of bilayer lipids at the same membrane fluidity.

It is striking that membranes made from stiffer bolalipids showed a curvature-dependent bending modulus, which is a clear signature that bolalipid membranes exhibit plastic behavior during membrane reshaping. Another marked difference between bilayer and flexible bolalipid membranes is that the Gaussian bending rigidity is almost twice as large as that of bilayer lipids and the ratio of the Gaussian rigidity to the bending modulus is around $-1/2$. It is not obvious how the Gaussian bending modulus would behave upon increasing bolalipid stiffness ($k_{\text{bola}} > 0$), or how to measure it due to the coupling between curvature and rigidity in bolalipid membranes. In any case, our results indicate that membrane remodelling, such as membrane fission during membrane traffic, is much more difficult in bolalipid membranes [34]. It is tempting to speculate that this is one of the reasons why eukaryotes use bilayer membranes, enabling dynamic membrane remodelling and trafficking.

It is interesting to draw a parallel between monolayer membranes made of stiff bolalipid molecules and macroscopic membranes composed of rigid straight colloidal particles, which are geometrically similar, but living at different scales [41]. It has been found that colloidal membranes at these macroscopic scales follow the standard Helfrich theory for bilayer membranes [42], with rigidity that is three orders of magnitude higher than those of lipid bilayers [43]. In this case the tilt modulus was not pertinent, likely due to macroscopic system sizes. Similarly, we expect that the bending rigidity can be determined from membrane fluctuations independently of the tilt modulus for bolalipid membranes if they are prepared at similar relative sizes. Beyond quantitative differences, the comparison shows that monolayer membranes follow the same physics across many orders of magnitudes. However, when considering subcellular scales and the formation of high curvature, as in vesicle budding, the tilt modulus and thus the substructure of the membrane is expected to matter.

We found that membranes formed of a mixture of bilayer and bolalipids, similar to archaeal membranes, function as a composite liquid membrane that softens when adding bilayer lipids. However, while in our simulations the bending rigidity monotonically decreases with bilayer fraction, previous experiments of mixture membranes of bilayer and bolalipids with cyclopentane rings suggested that the bending rigidity non-monotonically depends on the fraction of the membrane made up of bilayer lipids [18]. It remains to be determined whether the result is due to the specific lipids used, the resulting mismatch in lengths between two stacked bilayer lipids and a straight bolalipid, the experimental conditions or to non-linear effects such as the formation of lipid domains that soften the membrane with increasing bolalipid content. The same experiments reported that membranes consisting solely of bolalipids are more rigid than bilayer membranes and non-fluctuating, which is in agreement with our high bending modulus simulation results for near-pure bolalipid mixture membranes.

To investigate how bolalipid membranes respond to changing membrane curvature, we performed simulations in which small cargo particles budded from flat membranes. We found that by enforcing curvature on bolalipid membranes, the fraction of U-shaped bolalipids increased around the cargo bud, especially in the outer membrane layer and hence softened the membrane.

As another mechanism to release curvature stress we observed the formation of membrane pores, which could be mended by adding small amounts of bilayer lipids, similar to the mixture membranes that are found in archaea [37 [↗](#)]. Our results suggest that enforcing membrane bending can soften bolalipid membranes locally by increasing the number of U-shapes, rendering the membrane a mechanically switchable material where large curvature decreases stiffness.

Taken together, our results show how membranes which are mixtures of bilayer and bolalipids maintain cell integrity at high temperatures, while also undergoing leak free membrane bending. This suggest that archaeal membranes can balance opposing needs when adapting to extreme environmental conditions. Beyond understanding membrane properties and reshaping across the tree of life, these results pave the way for synthetic bolalipid membranes and bolalipid membrane containers with new and exciting material properties.

Data availability

The simulation input files and codes are freely available at [44 [↗](#)].

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Additional information

Author contributions

MA performed computer simulations and analysis and wrote the manuscript. FF performed patch closing simulations, contributed to computer simulations and analysis, and wrote the manuscript. XJ contributed to budding simulations and the manuscript. BB co-supervised the project and edited the manuscript. AŠ conceived the study, supervised the project and edited the manuscript.

Additional files

Movie S1. [↗](#)

Movie S2a. [↗](#)

Movie S2b. [↗](#)

Movie S2c. [↗](#)

Movie S3. [↗](#)

Movie S4. [↗](#)

Movie S5a. [↗](#)

Movie S5b. [↗](#)

Movie S5c. [↗](#)

Movie S5d. [↗](#)

Movie S6. [↗](#)

Supplemental information. [↗](#)

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Reviewer #1 (Public review):

Summary:

Amaral et al. presents a study investigating the mesoscale modelling and dynamics of bolalipids.

Strengths:

The figures in this paper are exceptional. Both those to outline and introduce the lipid types, but also the quality and resolution of the plots. The data held within also appears to be outstanding and of significant (hopefully) general interest.

Weaknesses:

In the introduction, I would like to have read more specifics on the biological role of bolalipids. Archaea are mentioned, but this kingdom is huge - there must be specific species that can be discussed where bolalipids are integral to archaeal life. The authors should go beyond 'extremophiles'. In short, they should unpack why the general audience should be interested in these lipids, within a subset of organisms that are often forgotten about.

Reviewer #2 (Public review):

Summary:

The authors aimed to understand the biophysical properties of archeal membranes made of bolalipids. Bacterial and eukaryotic membranes are made of lipids that self-assemble into bilayers. Archea, instead, use bolalipids, lipids that have two headgroups and can span the entire bilayer. The authors wanted to determine if the unique characteristics of archaea, which are often extremophiles, are in part due to the fact that their membranes contain bolalipids.

The authors develop a minimal computational model to compare the biophysics of bilayers made of lipids, bolalipids, and mixtures of the two. Their model enables them to determine essential parameters such as bilayer phase diagrams, mechanical moduli, and the bilayer behavior upon cargo inclusion and remodeling.

The author demonstrates that bolalipid bilayers behave as binary mixtures, containing bolalipids organized either in a straight conformation, spanning the entire bilayer, or in a u-shaped one, confined to a single leaflet. This dynamic mixture allows bolalipid bilayers to be very sturdy but also provides remodeling. However, remodeling is energetically more expensive than with standard lipids. The authors speculate that this might be why lipids were more abundant in the evolutionary process.

Strengths:

This is a wonderful paper, a very fine piece of scholarship. It is interesting from the point of view of biology, biophysics, and material science. The authors mastered the modeling and analysis of these complex systems. The evidence for their findings is really strong and complete. The paper is written superbly, the language is precise and the reading experience is very pleasant. The plots are very well-thought-out.

Weaknesses:

I would not talk about weaknesses, because this is really a nice paper. If I really had to find one, I would have liked to see some clear predictions of the model expressed in such a way that experimentalists could design validation experiments.

<https://doi.org/10.7554/eLife.105432.1.sa1>

Reviewer #3 (Public review):

Summary:

The authors have studied the mechanics of bolalipid and archaeal mixed-lipid membranes via comprehensive molecular dynamics simulations. The Cooke-Deserno 3-bead-per-lipid model is extended to bolalipids with 6 beads. Phase diagrams, bending rigidity, mechanical stability of curved membranes, and cargo uptake are studied. Effects such as the formation of U-shaped bolalipids, pore formation in highly curved regions, and changes in membrane rigidity are studied and discussed. The main aim has been to show how the mixture of bolalipids and regular bilayer lipids in archaeal membrane models enhances the fluidity and stability of these membranes.

Strengths:

The authors have presented a wide range of simulation results for different membrane conditions and conformations. For the most part, the analyses and their results are presented clearly and concisely. Figures, supplementary information, and movies very well present what has been studied. The manuscript is well-written and is easy to follow.

Major issues:

The Cooke-Deserno model, while very powerful for biophysical analysis of membranes at the mesoscale, is very much void of chemical information. It is parameterized such that it is good in producing fluid membranes and predicting values for bending rigidity, compressibility, and even thermal expansion coefficient falling in the accepted range of values for bilayer membranes. But it still represents a generic membrane. Now, the authors have suggested a similar model for the archaeal bolalipids, which have chemically different lipids (the presence of cyclopentane rings for one), and there is no good justification for using the same pairwise interactions between their representative beads in the coarse-grained model. This does not necessarily diminish the worth of all the authors' analyses. What is at risk here is the confusion between "what we observe this model of bolalipid- or mixed-membranes do" and "how real bolalipid-containing archaeal membranes behave at these mechanical and thermal conditions."

Another more specific, major issue has to do with using the Hamm-Kozlov model for fitting the power spectrum of thermal undulations. The $1/q^2$ term can very well be attributed to membrane tension. While a barostat is indeed used, have the authors made absolutely sure that the deviation from $1/q^4$ behavior does not correspond to lateral tension? I got more worried when I noticed in the SI that the simulations had been done with combined "fix langevin" and "fix nph" LAMMPS commands. This combination does not result in a proper isothermal-isobaric ensemble. The importance of tilt terms for bolalipids is indeed very interesting, but I believe more care is needed to establish that.

This issue is reinforced when considering Figure 3B. These results suggest that increasing the fraction of regular lipids increases the tilt modulus, with the maximum value achieved for a normal Cooke-Deserno bilayer void of bolalipids. But this is contradictory. For these bilayers, we don't need the tilt modulus in the first place.

Also, from the SI, I gathered that the authors have neglected the longest wavelength mode because it is not equilibrated. If this is indeed the case, it is a dangerous thing to do, because with a small membrane patch, this mode can very well change the general trend of the power spectrum. As a lot of other analyses in the manuscript rely on these measurements, I believe more elaboration is in order.

The authors have found that "there is a strong dependency of the bending rigidity on the membrane mean curvature of stiffer bolalipids." The effect is negative, with the membrane becoming less stiff at higher mean curvatures. Why is that? I would assume that with more flexible bolalipids, the possibility of reorganization into U-shaped chains should affect the bending rigidity more (as Figure 2E suggests). While for a stiff bolalipid, not much would change if you increase the mean curvature. This should be either a tilt effect, or have to do with asymmetry between the leaflets. But on the other hand, the tilt modulus is shown to decrease with increasing bolalipid rigidity. The authors get back to this issue only on page 10, when they consider U-shaped lipids in the inner and outer leaflets and write, "this suggested that an additional membrane-curving mechanism must be involved." But then again, in the Discussion, the authors write, "It is striking that membranes made from stiffer bolalipids showed a curvature-dependent bending modulus, which is a clear signature that bolalipid membranes exhibit plastic behavior during membrane reshaping," adding to the confusion.

This issue is repeated when the authors study nanoparticle uptake. They write: "to reconcile these seemingly conflicting observations we reason that the bending rigidity, similar to

Figure 2F, is not constant but softens upon increasing membrane curvature, due to dynamic change in the ratio between bolalipids in straight and U-shaped conformation. Hence, bolalipid membranes show stroking plastic behavior as they soften during reshaping." But the softening effect that they refer to, as shown in Figure 4B, occurs for very stiff bolalipids, for which not much switching to U-shaped conformation should occur.

Another major issue is with what the authors refer to as the "effective temperature". While plotting phase diagrams for kT/ϵ value is absolutely valid, I'm not a fan of calling this effective temperature. It is a dimensionless quantity that scales linearly with temperature, but is not a temperature. It is usually called a "reduced temperature". Then the authors refer to their findings as studying the stability of archaeal membranes at high temperatures. I have to disagree because ϵ is not the only potential parameter in the simulations (there are at least space exclusion and angle-bending stiffnesses) so one cannot identify changing ϵ with changing the global simulation temperature. This only works when you have one potential parameter, like an LJ gas.

Minor issues:

As the authors have noted, the fact that the membrane curvature can change the ratio of U-shaped to straight bolalipids would render the curvature elasticity non-linear (though the term "plastic" should not be used, as this is still structurally reversible when the stress is removed. Technically, it is hypoelastic behavior, possibly with hysteresis.) With this in mind, when the authors use essentially linear elastic models for fluctuation analysis, they should make a comparison of maximum curvatures occurring in simulations with a range that causes significant changes in bolalipid conformational ratios.

The Introduction section of the manuscript is written with a biochemical approach, with very minor attention to the simulation works on this system. Some molecular dynamics works are only cited as existing previous work, without mentioning what has already been studied in archaeal membranes. While some information, like the binding of ESCRT proteins to archaeal membranes, though interesting, helps little to place the study within the discipline. The Introduction should be revised to show what has already been studied with simulations (as the authors mention in the Discussion) and how the presented research complements it.

The authors have been a bit loose with using the term "stability". I'd like to see the distinction in each case, as in "chemical/thermal/mechanical/conformational stability".

In the original Cooke-Deserno model, a so-called "poorman's angle-bending term" is used, which is essentially a bond-stretching term between the first and third particle. However, I notice the authors using the full harmonic angle-bending potential. This should be mentioned.

The analysis of energy of U-shaped lipids with the linear model $E = c_0 + c_1 * k_{bola}$ is indeed very interesting. I am curious, can this also be corroborated with mean energy measurements? The minor issue is calling the source of the favorability of U-shaped lipids "entropic", while clearly an energetic contribution is found. The two conformations, for example, might differ in the interactions with the neighboring lipids.

The authors write in the Discussion, "In any case, our results indicate that membrane remodelling, such as membrane fission during membrane traffic, is much more difficult in bolalipid membranes [34]." Firstly, I'm not sure if studying the dependence of budding behavior on adhesion energy with nanoparticles is enough to make claims about membrane fission. Secondly, why is the 2015 paper by Markus Deserno cited here?

In the SI, where the measurement of the diffusion coefficient is discussed, the expression for D is missing the power 2 of displacement.

Where cargo uptake is discussed, the term "adsorption energy" is used. I think the more appropriate term would be "adhesion energy".

Typos:

Page 1, paragraph 2: Adaption → Adaptation.

Page 10, paragraph 1: Stroking → Striking.

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